

Hough Line Detection

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Preface

This tutorial is applicable to firmware version: CanMV_K230_YAHBOOM_micropython_v1.8-13. Download from Jianan official website https://download.kendryte.com/developer/releases/canmv_k230_micropython/daily_build/

 **K230 MicroPython 镜像下载**

CI/CD 更新时间: 2026-07-28 01:29:59

选择开发板

 CanMV K230 V1.0/V1.1

 CanMV K230 V3.0

 庐山派 K230

 庐山派 Lite K230D

 **Yahboom K230**

 01Studio K230

 01Studio K230 Emmc

 Hiwonder K230

 WonderMK K230

 GT6700 K230

 **Yahboom K230**
MicroPython Build for Yahboom K230

当前下载镜像:
[CanMV_K230_YAHBOOM_micropython_v1.8-28-g9b151af_nncase_v2.11.0.img.gz](#)

最新每日构建版本
Daily-0728
编译日期: 2026-07-28

MD5: 66f0622fc6b8d22e55f552a40458d2fa
立即下载

极简 USB 烧录指南 (推荐)

⚠ 镜像仅可烧录到对应的开发板, 请务必选择与当前板型完全匹配的镜像, 不能混烧其他板型的镜像。否则可能导致硬件损坏!!!

1 准备工具
下载并打开 [K230BurningTool](#) 烧录工具, 并加载固件。

2 进入烧录模式 (根据硬件情况选择)

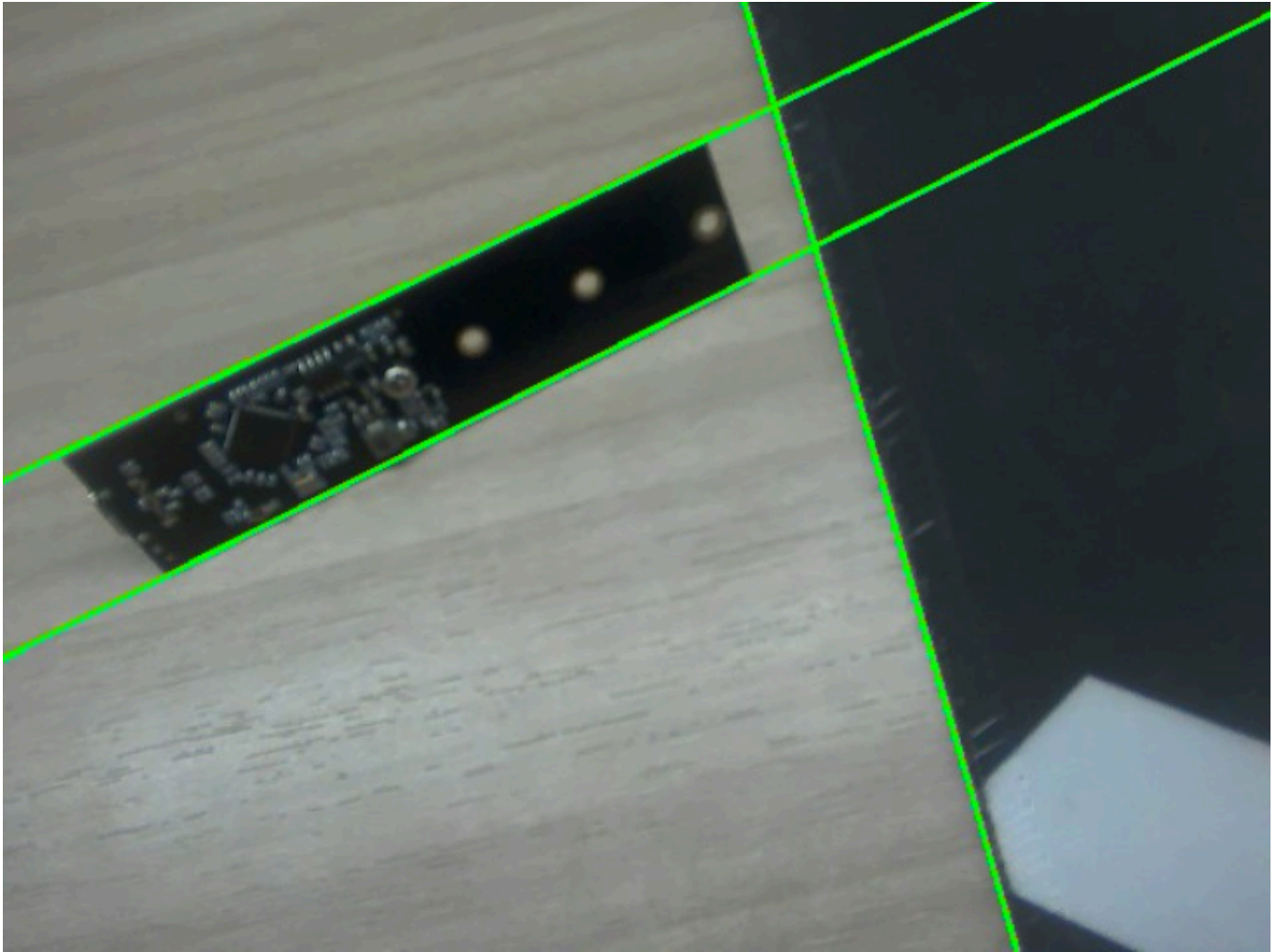
☒ 有 BOOT 键的板子

☐ 无 BOOT 键的板子

Example Results

Run the example code for this section [06.camera_hough_lines.py](#).

This section demonstrates standard Hough line and probabilistic Hough line segment detection on a 320x240 grayscale edge map, then maps the detection results back to the 640x480 original image for display.



Principle Explanation

Standard Hough Line

The standard Hough transform uses polar coordinates to represent lines:

$$x \cdot \cos\theta + y \cdot \sin\theta = \rho$$

Each edge point votes in the ρ - θ parameter space, and candidates exceeding the vote threshold are returned as lines. The return values are ρ and θ , which need to be converted to two endpoints on the image.

Probabilistic Hough Line Segment

HoughLinesP() directly returns (x1, y1, x2, y2) line segment endpoints, and limits results through minimum line length and maximum gap. It is more suitable for drawing finite line segments in actual images.

Code Overview

Initialize Camera and Display Module

```
W, H = 640, 480
sensor = Sensor(width=1280, height=960, fps=90)
sensor.reset()
sensor.set_framesize(width=W, height=H)
sensor.set_pixformat(Sensor.RGB888)
Display.init(Display.ST7701, width=640, height=480, to_ide=True)
sensor.run()
time.sleep(0.5)
```

Set Detection Mode

```
mode = 0
```

- mode=0: Standard HoughLines(), source code default.
- mode=1: Probabilistic HoughLinesP().

Downsample and Extract Edges

```
small = cv2.resize(img_np, (320, 240))
gray = cv2.cvtColor(small, cv2.COLOR_BGR2GRAY)
edges = cv2.Canny(gray, 50, 150)
```

Canny lower threshold is 50, upper threshold is 150. Hough transform only processes edge maps, not directly processing color images.

Standard Hough Line

```
lines = cv2.HoughLines(edges, 1, np.pi / 180, 120)

if lines is not None:
    for i in range(lines.shape[0]):
        if i >= 30:
            break
        rho = lines[i, 0]
        theta = lines[i, 1]
        a = np.cos(theta)
        b = np.sin(theta)
        x0 = a * rho * 2
        y0 = b * rho * 2
        cv2.line(
            img_np,
            (int(x0 + 1000 * (-b)), int(y0 + 1000 * a)),
            (int(x0 - 1000 * (-b)), int(y0 - 1000 * a)),
            (0, 255, 0), 2
        )
```

Distance resolution is 1 pixel, angle resolution is 1 degree, and accumulator threshold is 120. x0, y0 multiplied by 2 maps half-resolution coordinates back to the original image.

Probabilistic Hough Line Segment

```
lines = cv2.HoughLinesP(
    edges, 1, np.pi / 180, 70,
    minLineLength=30,
    maxLineGap=8
)

if lines is not None:
    for i in range(lines.shape[0]):
        if i >= 30:
            break
        cv2.line(
            img_np,
            (int(lines[i, 0]) * 2, int(lines[i, 1]) * 2),
            (int(lines[i, 2]) * 2, int(lines[i, 3]) * 2),
            (0, 0, 255), 2
        )
```

Minimum line length 30 and maximum gap 8 are both pixel values on the 320x240 image.

Resource Release

```
finally:
    sensor.stop()
    Display.deinit()
    os.exitpoint(os.EXITPOINT_ENABLE_SLEEP)
    time.sleep_ms(100)
```

Parameter Adjustment Notes

- **Canny Thresholds:** Increase when edge noise is high, decrease when weak lines are missing. Stabilize the edge map first, then modify Hough parameters.
- **Standard Hough Threshold 120:** Higher returns fewer but stronger lines.
- **Probabilistic Hough Threshold 70:** Controls the minimum vote count for line segment candidates.
- **minLineLength:** Filters short lines; actual length on original image is approximately twice the current value.
- **maxLineGap:** Allows gaps on the same line; too large will connect unrelated line segments.
- **Result Upper Limit:** Source code draws at most 30 lines, but printed lines.shape[0] may be greater than 30. Pay attention when reading serial data.
- **Scene Constraints:** Use ROI or angle range filtering to suppress background strong lines such as table edges and screen borders.